Chapter 1

Basics

This chapter is intended to introduce some of the most preliminary notions of category theory, which will be minimally required for defining categorical semantics in this book.

If mathematics may be likened to a zebra, then set theory regards a zebra as a black horse bearing white stripes, whereas category theory regards it as a white horse bearing black stripes. The decisive difference, one might say, is that category theory treats even a wholly unstriped white horse as a zebra all the same.

More concretely: a category is a monoid of morphisms. In set theory, a morphism was defined by appealing to the elements of sets, from which properties such as being injective were derived. In category theory, by contrast, morphisms are a primitive notion: the elements of sets are not only left unmentioned but need not even exist. Properties such as being injective are, as we shall see in the next section, stated instead as predications on morphisms.

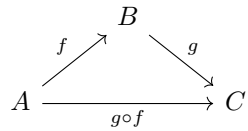
1.1 Categories

Definition 1.1.1 (Category)**.** A category $\mathcal{C}$ is specified by:

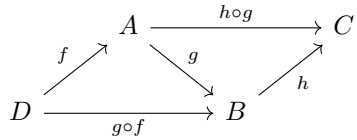
- a collection $ob\mathcal{C}$ of *objects*. An object is denoted by a capital letter such as A, B, C, ...
- a collection $mor\mathcal{C}$ of *morphisms* (also called *arrows*). A morphism is denoted by a lowercase letter such as f, g, h, ...
- two operations assigning to each morphism f its *domain* $dom(f)$ such that $dom(f) \in ob\mathcal{C}$ and its *codomain* $cod(f)$ such that $cod(f) \in ob\mathcal{C}$. (Notation $f : A \longrightarrow B$ where $A = dom(f)$ and $B = cod(f)$)

- an operation assigning to each pair of composable morphisms f and g their *composition* $g \circ f$ such that:

$$\begin{aligned} dom(g \circ f) &= dom(f) \\ cod(g \circ f) &= cod(g) \end{aligned} \qquad i.e.$$



$$(h \circ g) \circ f = h \circ (g \circ f) \qquad i.e.$$

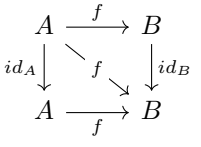


where f and g are *composable* if $cod(f) = dom(g)$.

- an operation assigning to each object A of $\mathcal{C}$ an *identity morphism* $id_A : A \longrightarrow A$ which satisfies the following axioms:

  For any f such that $dom(f) = A$, $\quad id_{cod(f)} \circ f = f$, $\quad f \circ id_{dom(f)} = f$

  i.e.



Remark 1.1.2. Axioms in category theory are equations about morphisms.

1.2 Examples of Categories

Example 1.2.1 (Sets)**. Set** is a category with objects all small sets, morphisms all functions between them with the usual composition of functions.

Example 1.2.2 (Monoids)**. Mon** is a category with objects all small monoids, morphisms all homomorphisms between them.

Example 1.2.3 (Topological Spaces)**. Top** is a category with objects topological spaces and morphisms continuous maps between them.

Example 1.2.4 (Matrices)**. Matr** is a category with objects all positive integers, morphisms $m \times n$ matrix A (regarded as a map $A : n \longrightarrow m$), and composition the usual matrix product.

Example 1.2.5 (Monoid)**.** A monoid $\langle M; \cdot; e\rangle$ is a category with object an arbitrary element $*$, morphisms the elements in M. For any $f, g \in M$, $dom(f) \stackrel{def}{=} *, cod(f) \stackrel{def}{=} *$ and $g \circ f \stackrel{def}{=} f \cdot g$. $id_* \stackrel{def}{=} e$.

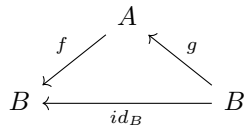
1.3 Commutative Diagrams

A *diagram* in a category $\mathcal{C}$ is an arrangement of vertices and edges, in which each vertex represents an object in $\mathcal{C}$ and each edge a morphism in $\mathcal{C}$. In particular, diagrams in which two or more paths merge (such as those shown in the previous section) play a special role.

Definition 1.3.1 (Commutative Diagram)**.** The diagram commutes if, taking any two vertices, the composite of morphisms along any path from the first vertex to the second is equal to the composite along any other path from the first to the second.

1.4 Isomorphisms

Definition 1.4.1 (Left and Right Inverses)**.** Given two morphisms $f : A \longrightarrow B$ and $g : B \longrightarrow A$ with $f \circ g = id_B$, g is a *right inverse* to f and f is a *left inverse* to g.



Definition 1.4.2 (Isomorphism)**.** A morphism $f : A \longrightarrow B$ is an *isomorphism* or *iso* (notation $f : A \xrightarrow{\sim} B$) if there is some $g : B \longrightarrow A$ for which $f \circ g = id_B$ and $g \circ f = id_A$. We shall say that g is an *inverse* for f and that f is an inverse for g.
Given objects A and B in $\mathcal{C}$, A is *isomorphic* to B (notation $A \cong B$) if such a mutually inverse pair of morphisms exists.

Theorem 1.4.3. For any morphism, its inverse is unique if it exists.

Proof. Suppose that two morphisms $g, h : B \longrightarrow A$ are inverses to a morphism $f : A \longrightarrow B$. Then each of the following equations holds.

$$\begin{array}{rclcrcl} g \circ f & = & id_A & \qquad & f \circ g & = & id_B \\ h \circ f & = & id_A & & f \circ h & = & id_B \end{array}$$

Thus we have $g = h$ as follows, which shows that f has a unique inverse.

$$\begin{array}{rcl} g & = & g \circ id_B \\ & = & g \circ (f \circ h) \\ & = & (g \circ f) \circ h \\ & = & id_A \circ h \\ & = & h \end{array}$$

□

Theorem 1.4.4. $\cong$ is an equivalence relation.

Proof. For any object A, id_A is iso since id_A itself is an inverse to it. Thus $A \cong A$ and isomorphism is reflexive. Suppose $A \cong B$, then a mutually inverse pair of morphisms exists thus $B \cong A$, so $\cong$ is symmetric. Suppose $A \cong B$ and $B \cong C$, then there exist mutually inverse pairs of morphisms, say $\langle f, g \rangle$ and $\langle h, k \rangle$, whose domains and codomains are as follows.

$$A \underset{g}{\overset{f}{\rightleftarrows}} B \underset{k}{\overset{h}{\rightleftarrows}} C$$

Then the following equations hold.

$$\begin{array}{rcl} & (g \circ k) \circ (h \circ f) & \\ = & g \circ (k \circ h) \circ f & \\ = & g \circ id_B \circ f & \\ = & g \circ f & \\ = & id_A & \end{array} \qquad \begin{array}{rcl} & (h \circ f) \circ (g \circ k) & \\ = & h \circ (f \circ g) \circ k & \\ = & h \circ id_B \circ k & \\ = & h \circ k & \\ = & id_C & \end{array}$$

Thus, $g \circ k$ and $h \circ f$ are a mutually inverse pair of morphisms and $A \cong C$. Therefore $\cong$ is transitive. □

Exercise 1.4.5. Given $f : A \longrightarrow B$, suppose that there is some $h : B \longrightarrow A$ with $h \circ f = id_A$ and some $g : B \longrightarrow A$ with $f \circ g = id_B$. Show that $g = h$, i.e. $g = f^{-1}$ is the unique inverse of f.

1.5 Monics and Epics

Definition 1.5.1 (Monic)**.** A morphism $f : A \longrightarrow B$ is *monic* iff for any object T and any pair of morphisms $g, h : T \longrightarrow A$, depicted as

$$T \underset{h}{\overset{g}{\rightrightarrows}} A \overset{f}{\rightarrowtail} B$$

$f \circ g = f \circ h$ implies $g = h$. As shown in the above diagram, we may write $f : A \rightarrowtail B$ to indicate that f is monic.

Remark 1.5.2. Note that the diagram in the above definition is not asserted to commute. Since a path consisting of a single edge has the morphism it represents as its composite, a diagram consisting of two parallel morphisms $g, h : T \longrightarrow A$ commutes if and only if $g = h$, which is the conclusion of the definition rather than its hypothesis. Diagrams of this kind, containing a pair of parallel morphisms that are not assumed to be equal, appear here and there in this book (the definition of epics below is another example). Such a diagram merely depicts the objects and morphisms under consideration; the equations that are assumed to hold are stated explicitly in the text.

Theorem 1.5.3. A composite of any two monics is monic.

Proof. Suppose that $f : A \longrightarrow B$ and $g : B \longrightarrow C$ are monic. For any object T and morphisms $h, k : T \longrightarrow A$ for which $g \circ f \circ h = g \circ f \circ k$, one has $f \circ h = f \circ k$ since g is monic. Then one also has $h = k$ since f is monic. This shows that the composite $g \circ f$ is monic.

$$T \underset{k}{\overset{h}{\rightrightarrows}} A \overset{f}{\rightarrowtail} B \overset{g}{\rightarrowtail} C$$

□

Exercise 1.5.4. Show that if a composite $g \circ f$ is monic, so is f. Is this also true of g?

Definition 1.5.5 (Epic)**.** A morphism $f : A \longrightarrow B$ is *epic* iff for any object T and any pair of morphisms $g, h : B \longrightarrow T$, depicted as

$$A \xrightarrow{f} B \underset{h}{\overset{g}{\rightrightarrows}} T$$

$g \circ f = h \circ f$ implies $g = h$. We may write $f : A \twoheadrightarrow B$ to indicate that f is epic.

Exercise 1.5.6. Prove that the composite of epics is epic.

Definition 1.5.7 (Split Epic)**.** A morphism is *split epic* if it has a right inverse.

Theorem 1.5.8. Every split epic is epic.

Proof. Suppose that $f : A \longrightarrow B$ is split epic and $g : B \longrightarrow A$ is a right inverse to f (i.e. $f \circ g = id_B$, recall Definition 1.4.1). Given two morphisms $h, k : B \longrightarrow T$ for which $k \circ f = h \circ f$, the following equations show that f is epic.

$$\begin{aligned} h &= h \circ id_B \\ &= h \circ (f \circ g) \\ &= (h \circ f) \circ g \\ &= (k \circ f) \circ g \\ &= k \circ (f \circ g) \\ &= k \circ id_B \\ &= k \end{aligned}$$

□

Theorem 1.5.9. A morphism is monic and split epic if and only if it is iso.

Proof. ($\Rightarrow$) Suppose $f : A \longrightarrow B$ is monic and split epic, and $g : B \longrightarrow A$ is its right inverse (i.e. $f \circ g = id_B$). Then $(f \circ g) \circ f = id_B \circ f = f$. On the other hand, $f \circ id_A = f$. Therefore $g \circ f = id_A$ since f is monic.

$$A \underset{id_A}{\overset{f}{\underset{g}{\rightrightarrows}}} A \overset{f}{\rightarrowtail} B$$

($\Leftarrow$) Suppose that $f : A \longrightarrow B$ is iso and $g : B \longrightarrow A$ its inverse. For any object T and morphisms $h, k : T \longrightarrow A$ such that $f \circ h = f \circ k$, one also has $g \circ f \circ h = g \circ f \circ k$. Since $g \circ f = id_A$, $h = k$ thus f is monic.

$$T \underset{k}{\overset{h}{\rightrightarrows}} A \underset{g}{\overset{f}{\leftrightarrows}} B$$

Since g is a right inverse of f, f is also split epic. □

1.6 Duality

Definition 1.6.1 (Opposite Category)**.** Given a category $\mathcal{C}$, the *opposite category* $\mathcal{C}^{op}$ is defined as follows:

- a collection $ob\mathcal{C}^{op}$ of objects of $\mathcal{C}^{op}$ is the same as $ob\mathcal{C}$.
- a collection $mor\mathcal{C}^{op}$ of morphisms of $\mathcal{C}^{op}$ is the same as $mor\mathcal{C}$. A morphism f in $\mathcal{C}$ is written as f^{op} as a morphism in $\mathcal{C}^{op}$.
- for any morphism f^{op} in $\mathcal{C}^{op}$, $dom(f^{op})$ in $\mathcal{C}^{op}$ is defined to be $cod(f)$ in $\mathcal{C}$, and $cod(f^{op})$ in $\mathcal{C}^{op}$ is defined to be $dom(f)$ in $\mathcal{C}$.
- for any morphisms $f^{op} : B \longrightarrow A$ and $g^{op} : C \longrightarrow B$ in $\mathcal{C}^{op}$, the composition of f and g in $\mathcal{C}^{op}$ is defined to be the morphism $f^{op} \circ g^{op} = (g \circ f)^{op} : C \longrightarrow A$ in $\mathcal{C}^{op}$. (Note that $f : A \longrightarrow B$ and $g : B \longrightarrow C$ in $\mathcal{C}$ are also composable, the composition of which is $g \circ f : A \longrightarrow C$.)

$\mathcal{C}$: $A \xrightarrow{f} B \xrightarrow{g} C$

$\mathcal{C}^{op}$: $C \xrightarrow{g^{op}} B \xrightarrow{f^{op}} A$

Definition 1.6.2 (Dual Statements)**.** The *dual* of any statement is formed by making the following replacements throughout the statement:

- "domain" by "codomain"
- "codomain" by "domain"
- "h is the composite of g and f" by "h is the composite of f and g"
- arrows and composites are reversed.

Definition 1.6.3 (Duality Principle)**.** The dual of each of the axioms for a category is also an axiom. Hence in any proof of a theorem about an arbitrary category from the axioms replacing each statement by its dual gives a valid proof.

Constructions shown in the subsequent sections give examples of duality.

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